 Name________________________

Three-Dimensional Vector Cross Products

Date________________ Period____

Find the cross product of the given vectors.

1) $\vec{a} \times \vec{b}$
$\vec{a} = \langle -9, 5, -3 \rangle$
$\vec{b} = \langle 2, 9, 2 \rangle$

2) $\overrightarrow{AB} \times \overrightarrow{CD}$
Given: $A = (-6, -2, 6)$ $B = (-5, -5, -3)$
$C = (8, 7, -6)$ $D = (-5, -4, -1)$

Find a vector that is perpendicular to the given vectors.

3) $\vec{u} = \langle -5, 6, 7 \rangle$
$\vec{v} = \langle 1, -9, 0 \rangle$

4) $\overrightarrow{RS}$ and $\overrightarrow{RT}$
Given: $R = (3, 7, 2)$ $S = (1, 7, -7)$
$T = (-1, -5, 6)$

Find the area of a triangle with the given vectors as two adjacent sides.

5) $\vec{a} = \langle 9, -2, 9 \rangle$
$\vec{b} = \langle 4, 9, -2 \rangle$

6) $\vec{u} = \langle 0, 2, -1 \rangle$
$\vec{v} = \langle -3, 6, 8 \rangle$